

Bhargav Singapuri

AI / Machine Learning Engineer | Seeking Summer 2026 Internship | Available May – August 2026
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PROFESSIONAL SUMMARY

AI/ML Engineer with production experience deploying LLM systems, REST APIs, and monitoring pipelines in healthcare. Re-architected Flask backend from 10% to 80% test coverage, integrated LangSmith for inference cost optimization (34% reduction), and built LLM summarization pipelines cutting turnaround from 2 weeks to 2 minutes. Strong in Python, Docker, CI/CD, and cloud infrastructure.

EDUCATION

Northeastern University <i>Master of Science in Artificial Intelligence GPA: 3.83</i>	Boston, MA <i>September 2024 – December 2026 (Expected)</i>
Nanyang Technological University <i>Bachelor of Science in Data Science and Artificial Intelligence GPA: 3.73</i>	Singapore <i>August 2019 – July 2023</i>

SKILLS

MLOps & Deployment	Docker, LangSmith, Terraform, Airflow, REST APIs, CI/CD Pipelines
ML/AI Frameworks	PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, LangChain, OpenCV, TransformerLens, SAELens
Data & Analytics	SQL (PostgreSQL, Snowflake), Pandas, NumPy, Airflow, Tableau, Excel (PowerQuery & Pivot Tables)
Cloud & Infrastructure	AWS, Google Cloud Platform (GCP), Azure
Programming & Tools	Python, Shell Scripting, Git, Linux/UNIX, CI/CD
Specialisations	Natural Language Processing (LLMs, RAG, Prompt Engineering, Agents), Computer Vision (CNNs, ViTs), Model Interpretability, Monte-Carlo Simulation, Data Visualization

EXPERIENCE

Biogen <i>Machine Learning Co-op</i>	Boston, MA <i>July 2025 – December 2025</i>
Backend Modernization: Re-architected Flask backend API using modular blueprints, reducing code coupling and improving maintainability; cut average feature integration time by 30% and increased unit test coverage from 10% → 80% Monitoring & Cost Optimization: Integrated LangSmith into internal AI apps to monitor token usage, benchmark responses, and optimize prompts — lowering inference costs 34% and raising stakeholder satisfaction by 16% Process Automation: Designed and deployed an LLM-powered summarization system that extracts key points from pTRS meeting minutes and auto-generates leadership-specific 1-page briefs, cutting turnaround from 2 weeks to 2 minutes Engineering Enablement: Authored and standardized an internal AI-assisted development playbook for Claude Code and OpenAI Codex, adopted by 10+ engineers, reducing prototyping time and accelerating delivery of features across 3 workflows Knowledge Deduplication: Implemented ML pipelines to detect redundant SOPs via semantic clustering, reducing internal documentation load by 10%	
United Overseas Bank <i>Data Engineer</i>	Singapore <i>July 2023 – August 2024</i>
Dashboarding: Spearheaded a Splunk dashboard visualizing TPS, latency, and HTTP statuses on 10M+ daily API records, cutting incident diagnosis time from 10 hours to 10 seconds Infrastructure Reliability: Collaborated with cross-functional teams to deliver database upgrades, ensuring minimal downtime and security of customer data	
Vertex Holdings <i>Data Science Intern</i>	Singapore <i>December 2021 – April 2023</i>
Data Pipeline Engineering: Developed Airflow ETL pipelines ingesting 1M+ relationship and investment records into Snowflake, creating a unified data layer that improved accessibility of investor intelligence for investment analysis and deal sourcing Investor Relationship Platform: Built a CRM that centralized data on investment and business-development opportunities, generating relationship graphs and a searchable knowledge base for investment managers to map investor networks and automatically log outreach, eliminating manual reconciliation and speeding up deal flow	

PROJECTS

Hidden in Plain Sight - Using Transliteration as an attack vector on Gemma 3 models to elicit harmful responses - Building mitigation tools using linear probes and SAEs	<i>January 2026 – Present</i>
Cross-coder Interpretability Benchmarking Cross-coders as part of quiver of arrows in a practitioners' toolkit on binary decision datasets	<i>January 2026 – Present</i>
Cancer Cell Detection - Fine-tuned Vision Transformers on lung-cell imagery using PyTorch Lightning; achieved 99% accuracy / 0.99 F1 - Quantified uncertainty with Monte-Carlo Dropout to ensure reliability in clinical inference scenarios	<i>January 2025 – May 2025</i>
JournalGPT Created a Retrieval-Augmented Generation (RAG) pipeline via Flowise, enabling semantic querying of personal journal entries with conversational tone-matching	<i>September 2024 - October 2024</i>